

FACULTAD DE INGENIERÍA
UNIVERSIDAD DE SAN CARLOS DE GUATEMALA

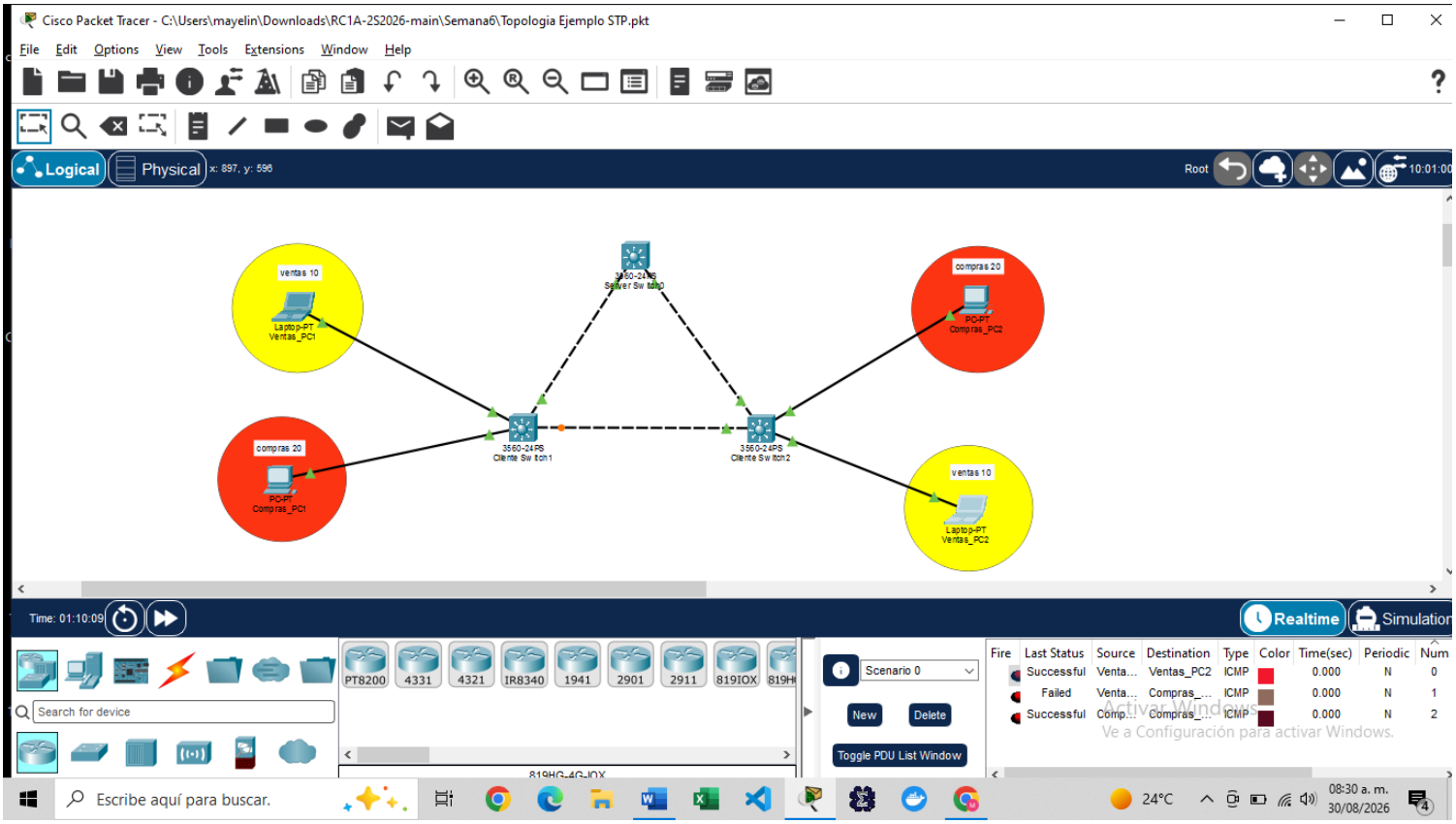
Universidad de San Carlos de Guatemala
Facultad de Ingeniería
Laboratorio redes de computadoras 1
Aux: Pablo Rodríguez

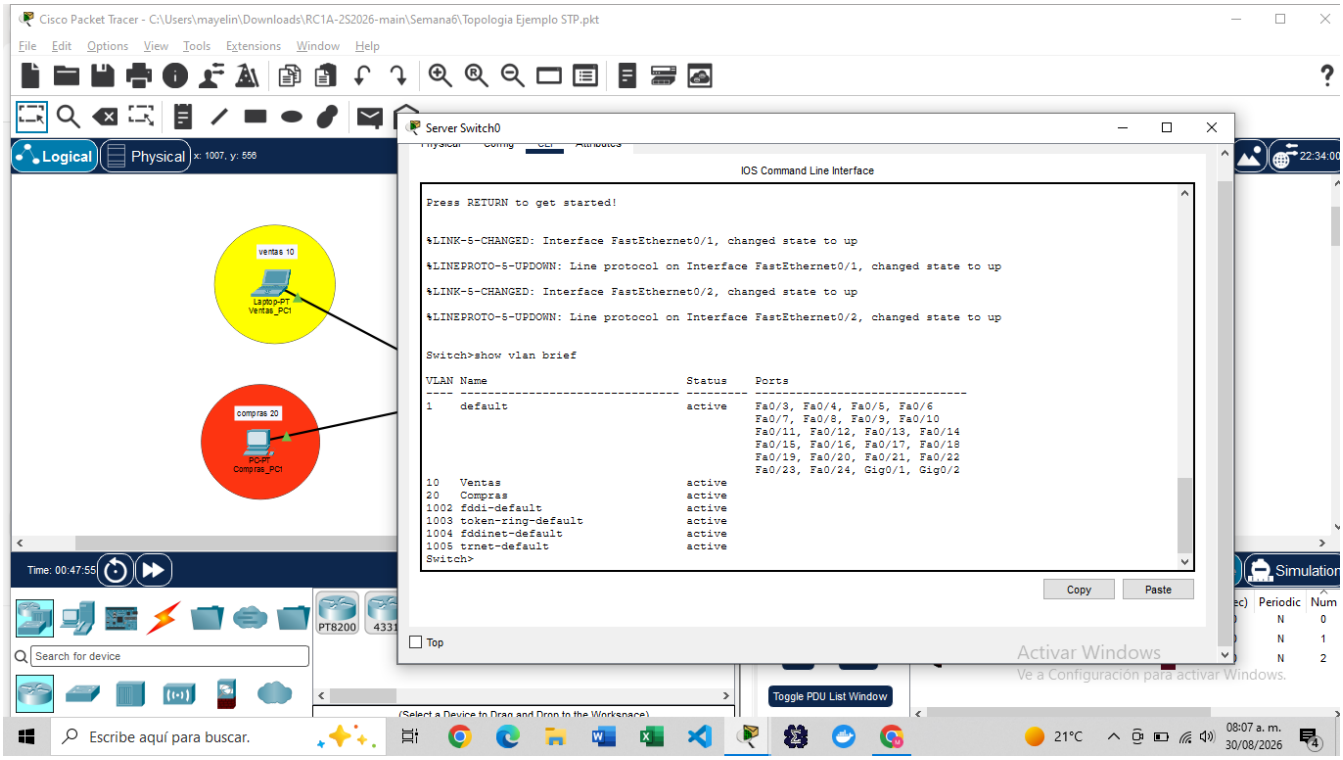
Actividad 3

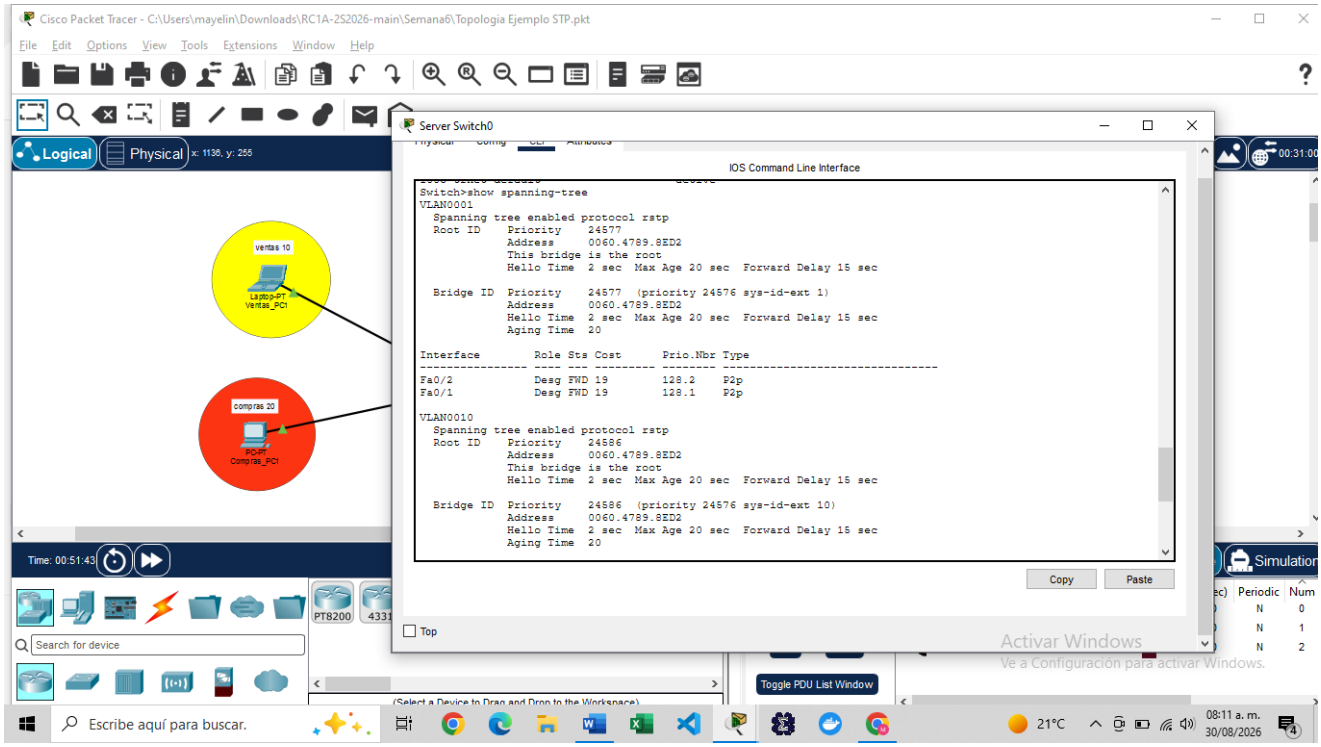
Nombre: Mayelin Lucía Quevedo Camey

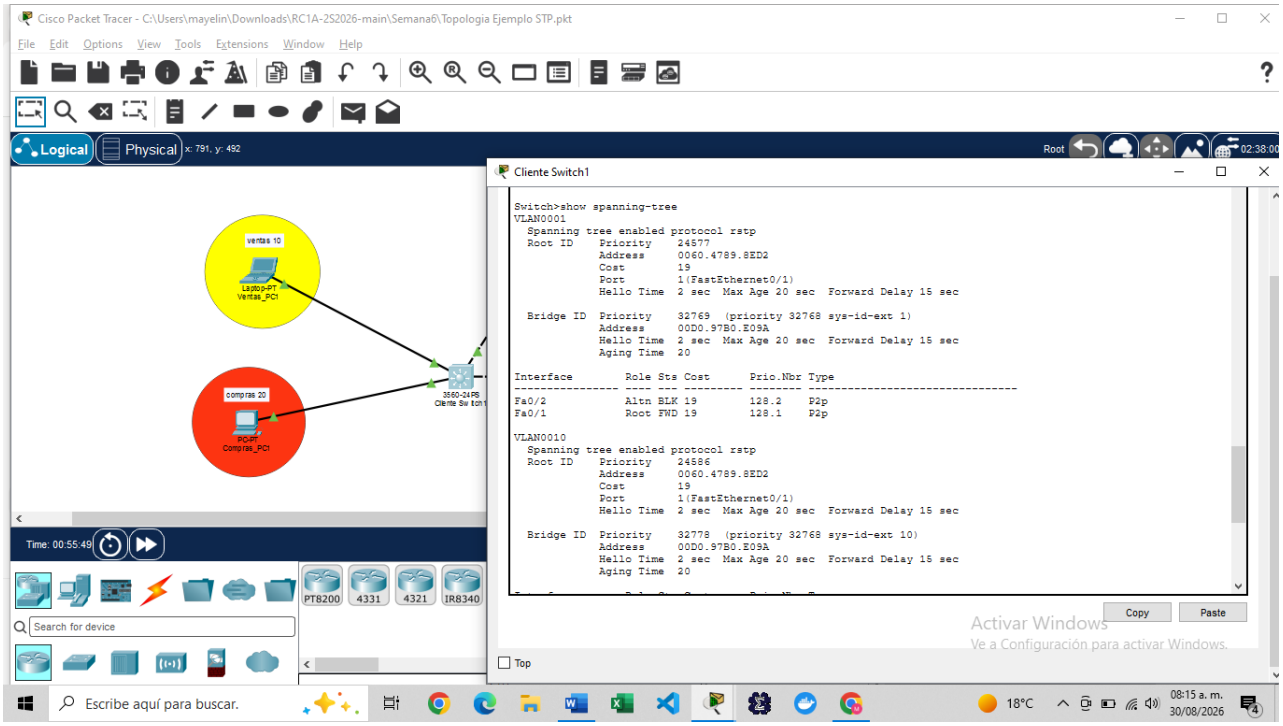
Carnet: 202307771

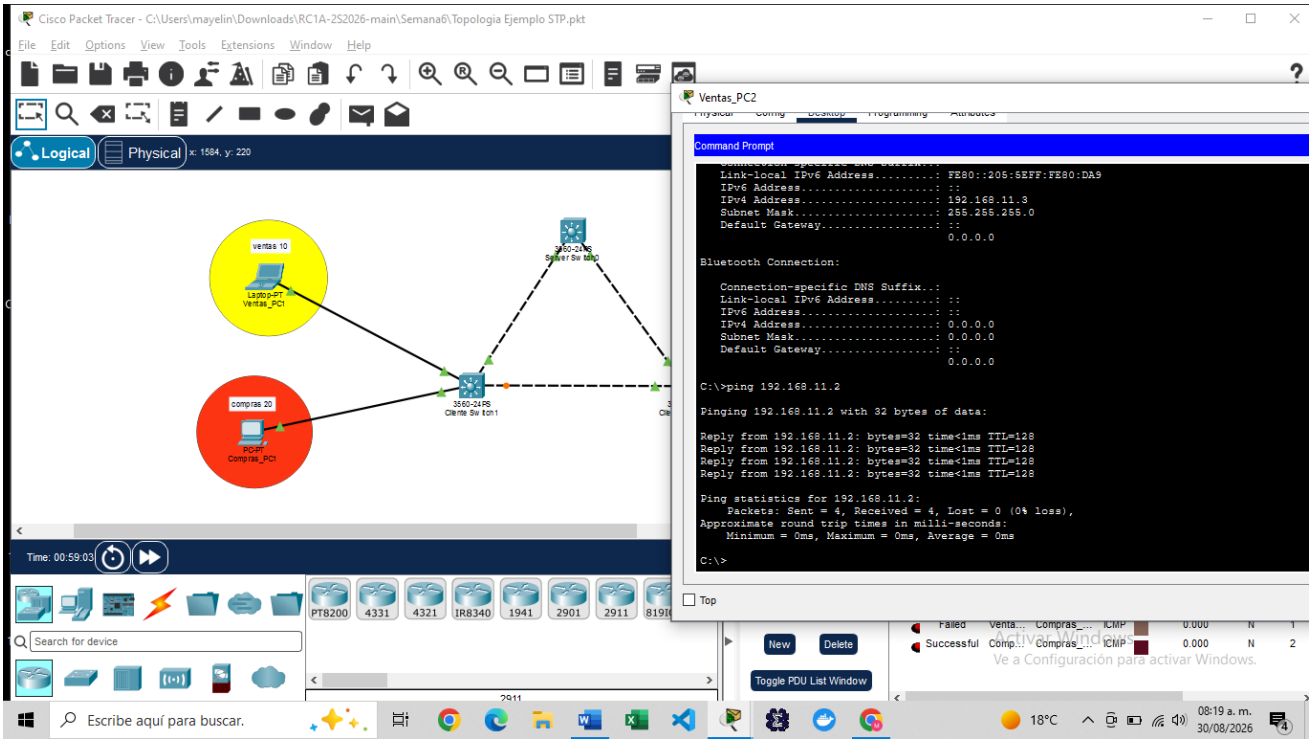
Guatemala 29 de agosto 2026


Cisco Packet Tracer - C:\Users\mayelin\Downloads\RC1A-2S2026-main\Semana6\Topologia Ejemplo STP.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
ventas 10
Laptop-PT
Ventas_PC1
compras 20
PC-PT
Compras_PC2
compras 20
PC-PT
Compras_PC1
ventas 10
Laptop-PT
Ventas_PC2
Time: 01:10:09
Realtime
Simulation
Search for device
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Successful
Failed
Successful
ICMP
0.000
Activar Windows
Ve a Configuración para activar Windows.
Escribe aquí para buscar.
24°C
08:30 a. m.
30/08/2026


Cisco Packet Tracer - C:\Users\mayelin\Downloads\RC1A-2S2026-main\Semana6\Topologia Ejemplo STP.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Server Switch0
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>show vlan brief
VLAN Name Status Ports
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/1, Gig0/2
10 Ventas active
20 Compras active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch>
Copy
Paste
Top
ventas 10
Laptop-PT Ventas_PC1
compras 20
PC-PT Compras_PC1
Time: 00:47:55
Simulation
Toggle PDU List Window
Activar Windows
Ve a Configuración para activar Windows.
Escribe aquí para buscar.
21°C
08:07 a. m.
30/08/2026


Cisco Packet Tracer - C:\Users\mayelin\Downloads\RC1A-2S2026-main\Semana6\Topologia Ejemplo STP.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Server Switch0
IOS Command Line Interface
Switch>show spanning-tree
VLAN0001
Spanning tree enabled protocol rstp
Root ID Priority 24577
Address 0060.4789.8ED2
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
Address 0060.4789.8ED2
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/1 Desg FWD 19 128.1 P2p
VLAN0010
Spanning tree enabled protocol rstp
Root ID Priority 24586
Address 0060.4789.8ED2
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 24586 (priority 24576 sys-id-ext 10)
Address 0060.4789.8ED2
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Copy
Paste
Top
ventas 10
Laptop-PT Ventas_PC1
compras 20
PC-PT Compras_PC1
Time: 00:51:43
Simulation
Toggle PDU List Window
Activar Windows
Ve a Configuración para activar Windows.
Escribe aquí para buscar.
21°C
08:11 a. m.
30/08/2026





- Con el comando show spanning-tree, identifiqué los roles y estados de los puertos en las diferentes VLAN configuradas.
- El switch0 fue Root Bridge, es decir, es el switch raíz
- El puerto Fa0/2 se encuentra en estado Blocking, debido a que existe un camino redundante. RSTP lo mantiene bloqueado para evitar la formación de bucles.
- Finalmente, se comprobó que RSTP permite mantener una red con enlaces redundantes sin generar bucles, seleccionando automáticamente los caminos principales y dejando otros como respaldo.
- EN EL PING SE OBSERVA QUE LA IP, ES 192.168.11.2, mi número de carnet es 202307771 terminación 1.